

University Intake Analysis

Your Name

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Introduction

This report examines programme intake across campuses and academic years.

Data preparation

The following task describes the required data-preparation work.

Task 1: Prepare the intake data

- Import the university intake dataset.
- Check missing values.
- Classify programs as UG or PG.
- Calculate campus-level totals.

Task 2: Analyse programme intake

Compare intake according to:

1. Academic year
2. Campus
3. UG and PG level
4. Programme duration

Produce a heatmap showing the final values.

```
library(tidyverse)
# Your R analysis
mtcars %>% head(5) %>% knitr::kable(format = "pipe")
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2

```
#include: false
#| echo: true
library(dplyr)
avgMPG <- mtcars %>% summarise(avg_mpg = mean(mpg))

bestCar <- mtcars %>% filter(mpg == max(mpg)) %>% rownames_to_column(var='car')
%>% select(car, mpg)

typst_content <- c(  '#import "typst-functions.typ": task-box',  "",
sprintf( '#task-box([Result generated by R])[Average MPG: *%.1f*. Highest-MPG
car: *%s*.]',  avgMPG, bestCar ))

writelines(typst_content, "generated-result.typ")
```

The following Typst content was generated by the preceding R calculation:

Result generated by R

Average MPG: **20.1**. Highest-MPG car: **Toyota Corolla**.

Result generated by R

Average MPG: **20.1**. Highest-MPG car: **33.9**.

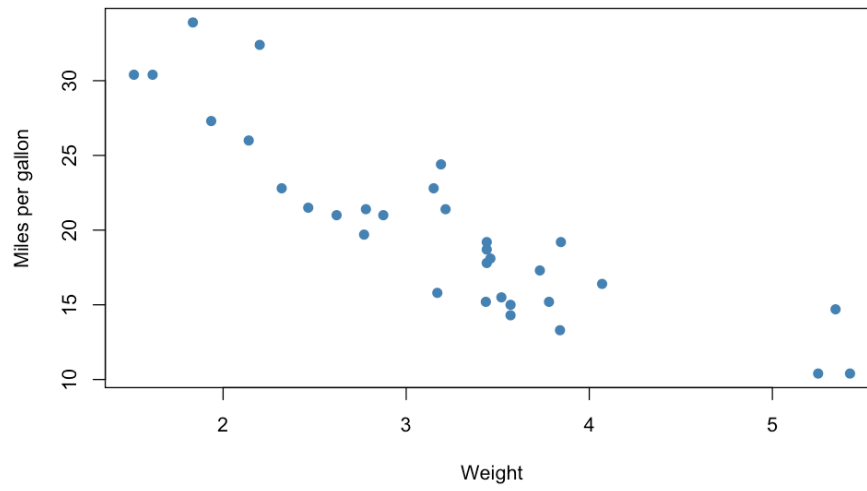


Figure 1: Relationship between vehicle weight and fuel efficiency